Chrome File Edit View History Bookmarks Profiles Tab Window Help

Sat 17 Oct 2:23 AM

edu.huaweicloud.com/intl/en-us/lab/experimentalStudy_povgunbkqg_2028_0_1792185250866

HUAWEI CLOUD Using NumPy Arrays

0% Time remaining : 00 : 50 : 35 End Exercise

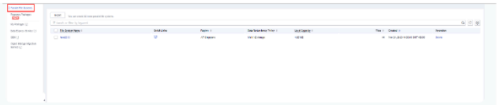
Tutorials Reports

Lab Account One-Click Login Credentials

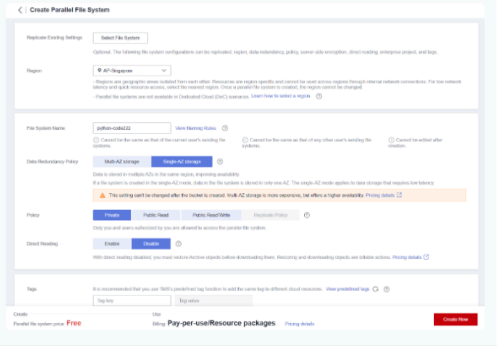
Account es_edu_sandbox_2260

Username Sandbox-user

Password **********



On the **Create Parallel File System** page, set parameters.



Steps Previous chapter Next chapter

Object Storage Service-C

console.huaweicloud.com/obs/?locale=en-us&agencyId=01a13bdb85...

Protection required for critical operations. To enable protection, go to Security Settings > Critical Operations > Operation Protection. Enable

HUAWEI CLOUD Console Search for... More EN es_edu_sandbox Sandbox-user

Parallel File System / Create Parallel File System

Create Parallel File System

General Configuration

Region

AP-Singapore

Once a parallel file system is created, its region **cannot** be changed. Resources are region-specific and they can be connected through an internal network within the same region. For lower latency and faster access, sele the region nearest where your parallel file system will be accessed. Learn how to select a region.

Parallel File System Settings

File System Name

python-code222hamza View Naming Rules

Domain name of a parallel file system:*python-code222hamza.obs.ap-southeast-3.myhuaweicloud.com*

Data Redundancy Policy

Multi-AZ storage Single-AZ storage

This setting **cannot be changed** after the parallel file system is created. Data in the parallel file system is stored across multiple availability zones (AZs) within the region. If one AZ becomes unavailable, the data remains accessible from the other AZs.

Multi-AZ storage can be covered by Standard (multi-AZ) storage packages. Pricing details

Block Public Access

Enabled settings: 4

If Block Public Access is enabled for a parallel file system, the performance will deteriorate (**Increased request duration**).
Enabling Block Public Access can help prevent data leaks and theft.

Show Settings If you require public access at certain levels, you can customize the settings below to suit your specific use cases.

Bucket Price: **Free**

Object Storage Service-...

Chrome File Edit View History Bookmarks Profiles Tab Window Help

Sat 17 Oct 2:32 AM

edu.huaweicloud.com/intl/en-us/lab/experimentalStudy_povgunbkqg_2028_0_1792185250866

HUAWEI CLOUD Using NumPy Arrays

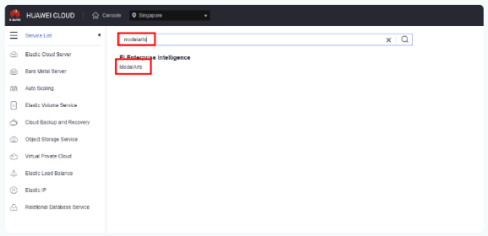
0% Time remaining : 00 : 42 : 10 End Exercise

Tutorials Reports

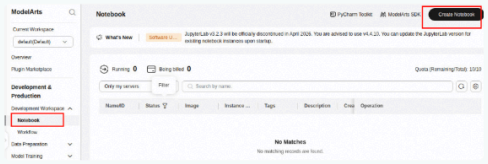
Lab Account One-Click Login Credentials

Step 1 Accessing the ModelArts Console

On the Huawei Cloud service list page, choose **ModelArts** to go to the ModelArts console.



In the navigation pane, choose **Development Workspace** > **Notebook** and click **Create**.



On the **Create Notebook** page, set parameters.

Steps Previous chapter Next chapter

console.huaweicloud.com/obs/?locale=en-us&agencyId=01a13bdb85...

Protection required for critical operations. To enable protection, go to Security Settings > Critical Operations > Operation Protection. Enable

HUAWEI CLOUD Console Search for... More EN es_edu_sandbox Sandbox-user

Parallel File System List / Files

python-code222hamza Standard | Single-AZ storage | AP-Singapore | Created Oct 16, 2026 21:25:50 GMT+00:00 Feedback Task Ce

Overview
Files
Metrics NEW
Basic Configurations
Tags
Permissions
Policy
ACL
Block Public Access
IAM Permissions
Data Security
CORS Rules
Data Management
Lifecycle Rules
Event Notification
Log Management
Log Storage

Files

Parallel File System (PFS) is a high-performance file system provided by OBS. It can be part of a complete big data solution where OBS serves as a unified data lake. Learn more

Files Fragments

You can use OBS Browser+ to move a file to any other folder in this parallel file system. For security reasons, files cannot be previewed online when you access them from a browser. To preview files online, see How Do I Preview Objects in OBS from My Browser?

Upload File Create Folder Delete More

Enter a file name prefix.

Name	Size	Last Modified	Operation
hamza-numpy	Not analyzed	--	Analyze Copy Path Delete More

Total Records: 1 Selected: 0 50 1

Object Storage Service-...

HUAWEI CLOUD Using NumPy Arrays 0% Time remaining : 00 : 32 : 51 End Exercise

Tutorials Reports

Lab Account One-Click Login Credentials

Refer to the following configuration and click **Next**.

Name: Enter **notebook-1221** in this practice.

Image: Choose **Public image** and select **pytorch_1.12.1-cuda_10.2-py_3.9.11-ubuntu_20.04-x86_64** (The specific image is not limited here. Please select x86 Python environment and system image.).

Resource Type: Select **Public resource pool**.

Flavor: Select **CPU: 2 vCPUs 8 GB**.

Retain default settings for other parameters.

Click **Submit**.The notebook instance will be created after a while.



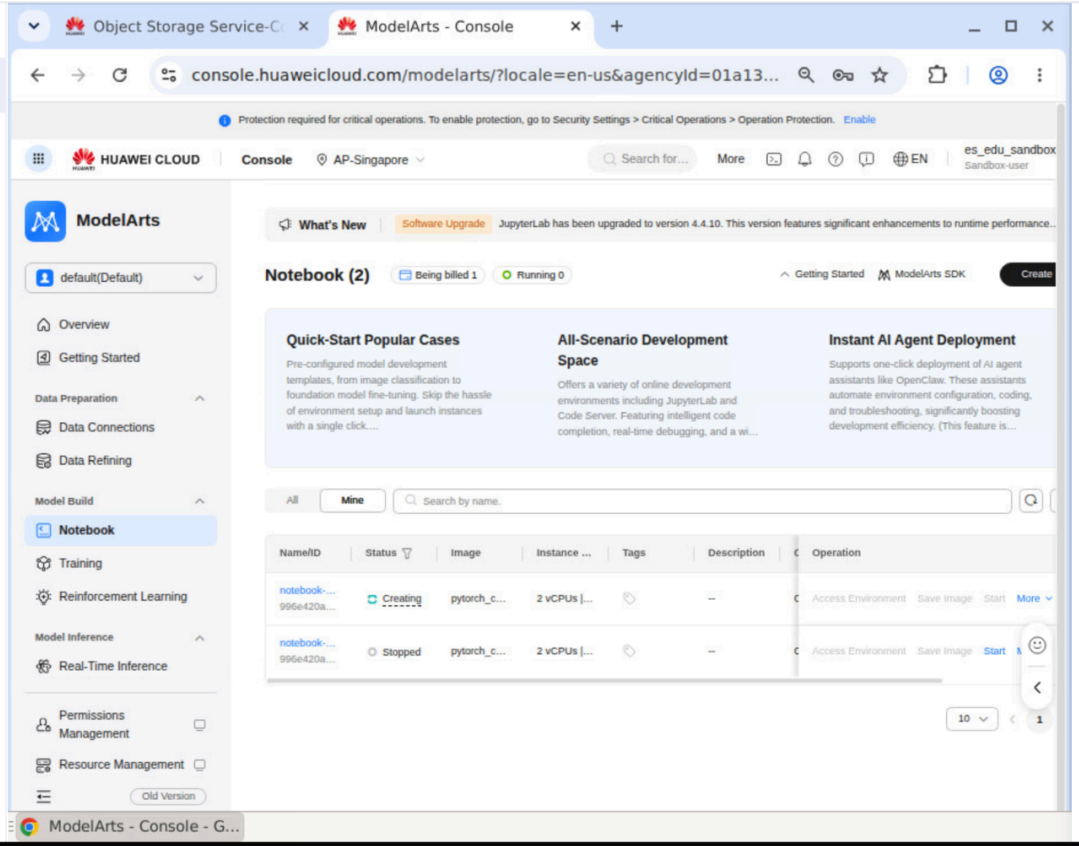
Task 3: Mounting the Notebook Instance to OBS

Step 1 Mounting the Notebook Instance

Click the created notebook instance. On the page that appears, choose the **Storage** tab and click **Mount Storage**.



Steps Previous chapter Next chapter



HUAWEI CLOUD Using NumPy Arrays

33% Time remaining : 00 : 26 : 36 End Exercise

Tutorials Reports

Lab Account One-Click Login Credentials

Step 4 Using the array() Function Provided by NumPy to Create a One-Dimensional Array

Enter and run the following code:

```
#Import the NumPy library. The following experi
import numpy as np
#Create a one-dimensional array
arr1 = np.array([1, 2, 3, 4])
print('Array Created:',arr1)
np.save("/data/input/task_02_arr1", arr1)
```

The following figure shows the result.



Step 5 Defining a NumPy Multi-dimensional Array

Steps Previous chapter Next chapter

authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...

File Edit View Run Kernel Git Tabs Settings Help Time Left:55mins

ModelArts Launcher numpy.ipynb

Code

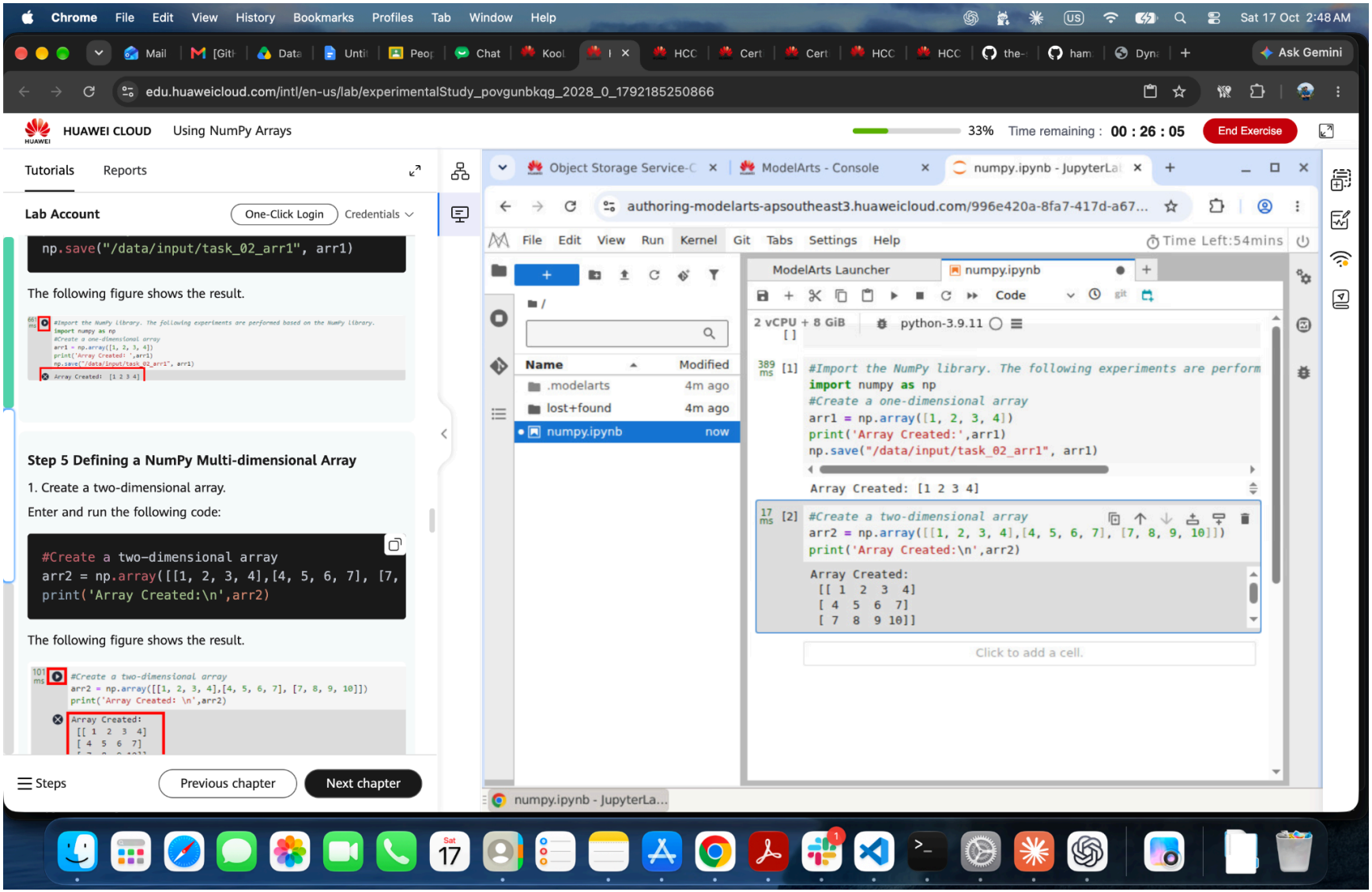
2 vCPU + 8 GiB python-3.9.11

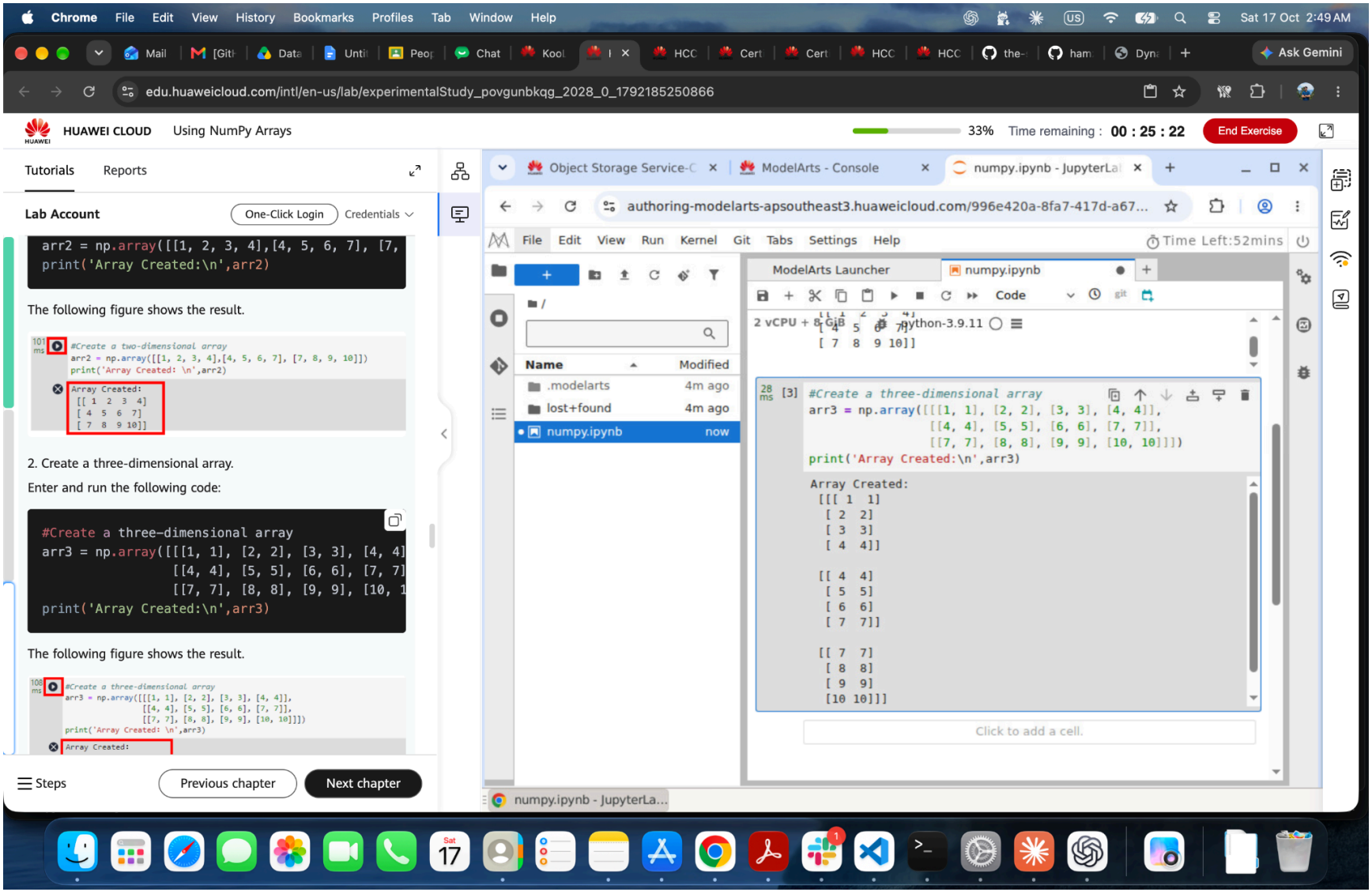
```
#Import the NumPy library. The following experiments are perform
import numpy as np
#Create a one-dimensional array
arr1 = np.array([1, 2, 3, 4])
print('Array Created:',arr1)
np.save("/data/input/task_02_arr1", arr1)
```

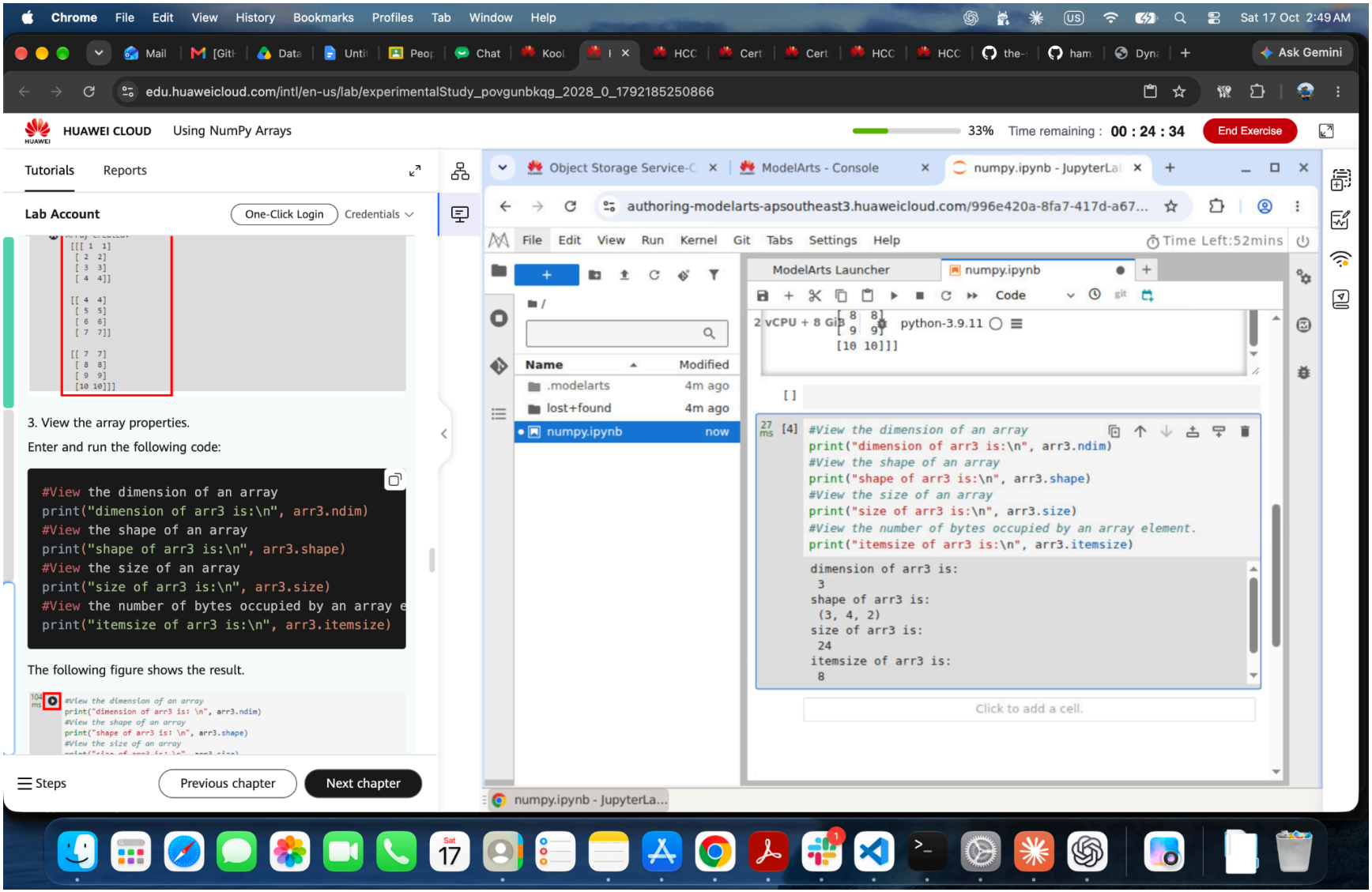
Array Created: [1 2 3 4]

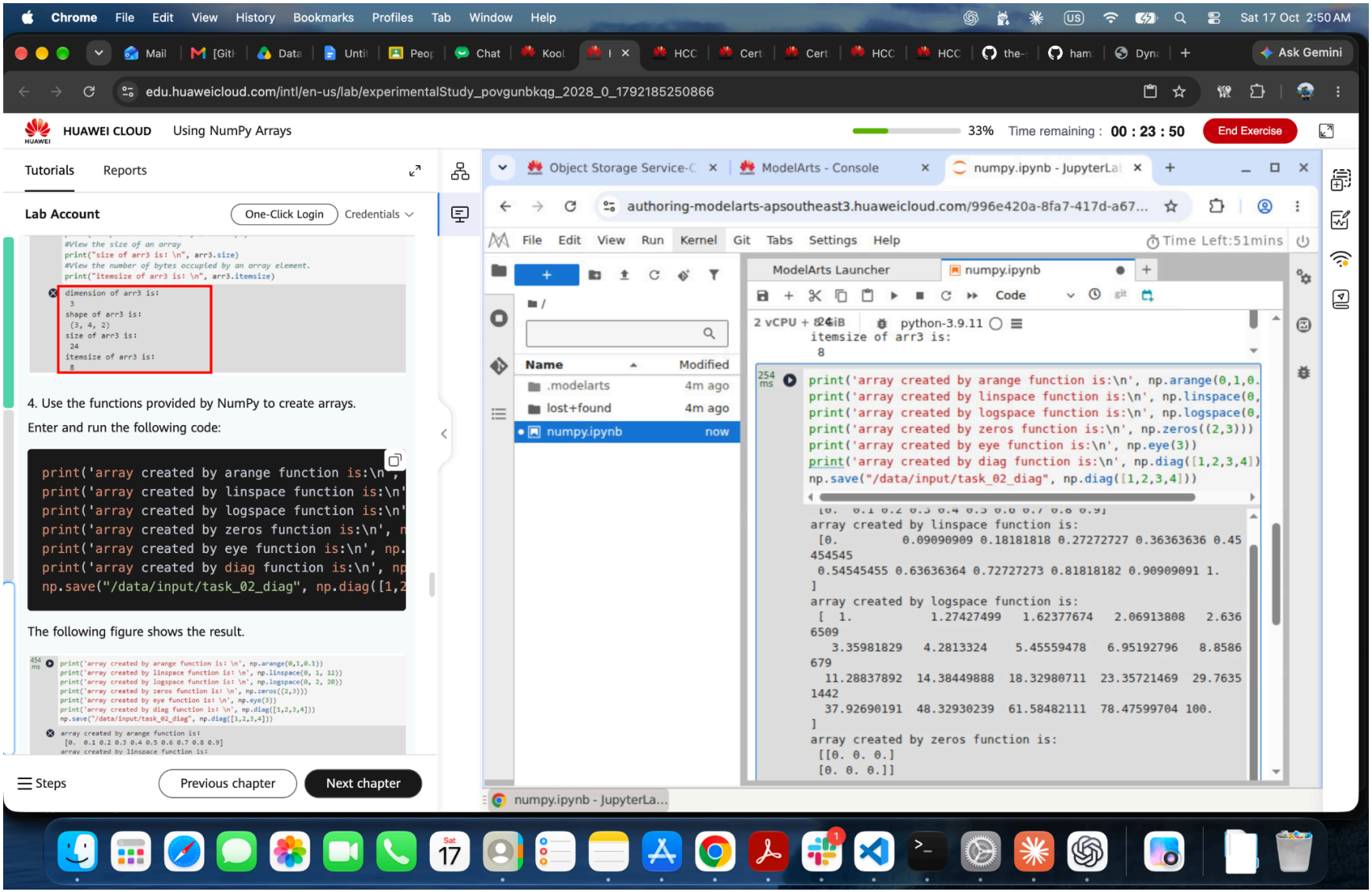
Click to add a cell.

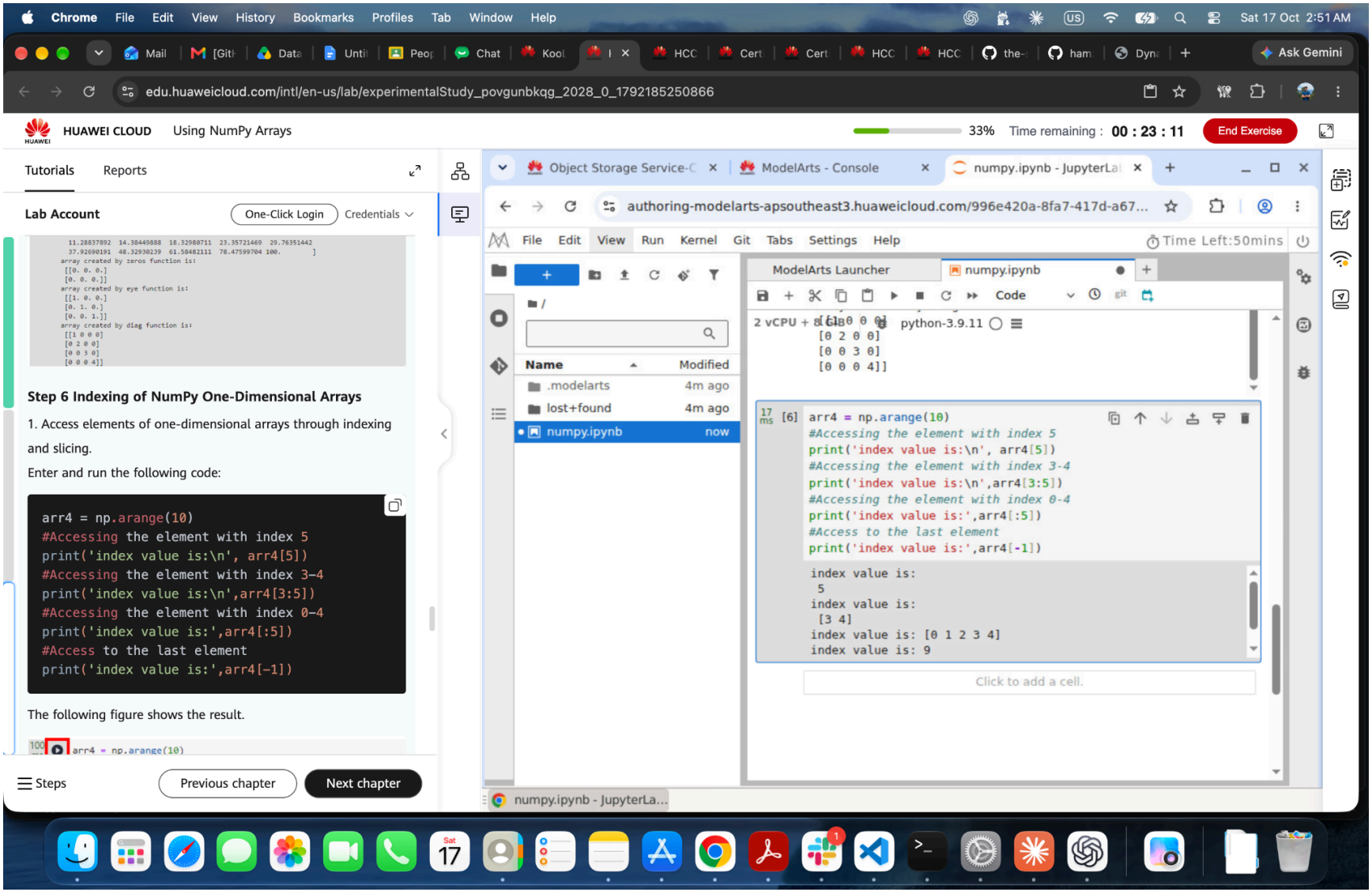
Name	Modified
.modelarts	4m ago
lost+found	4m ago
numpy.ipynb	now

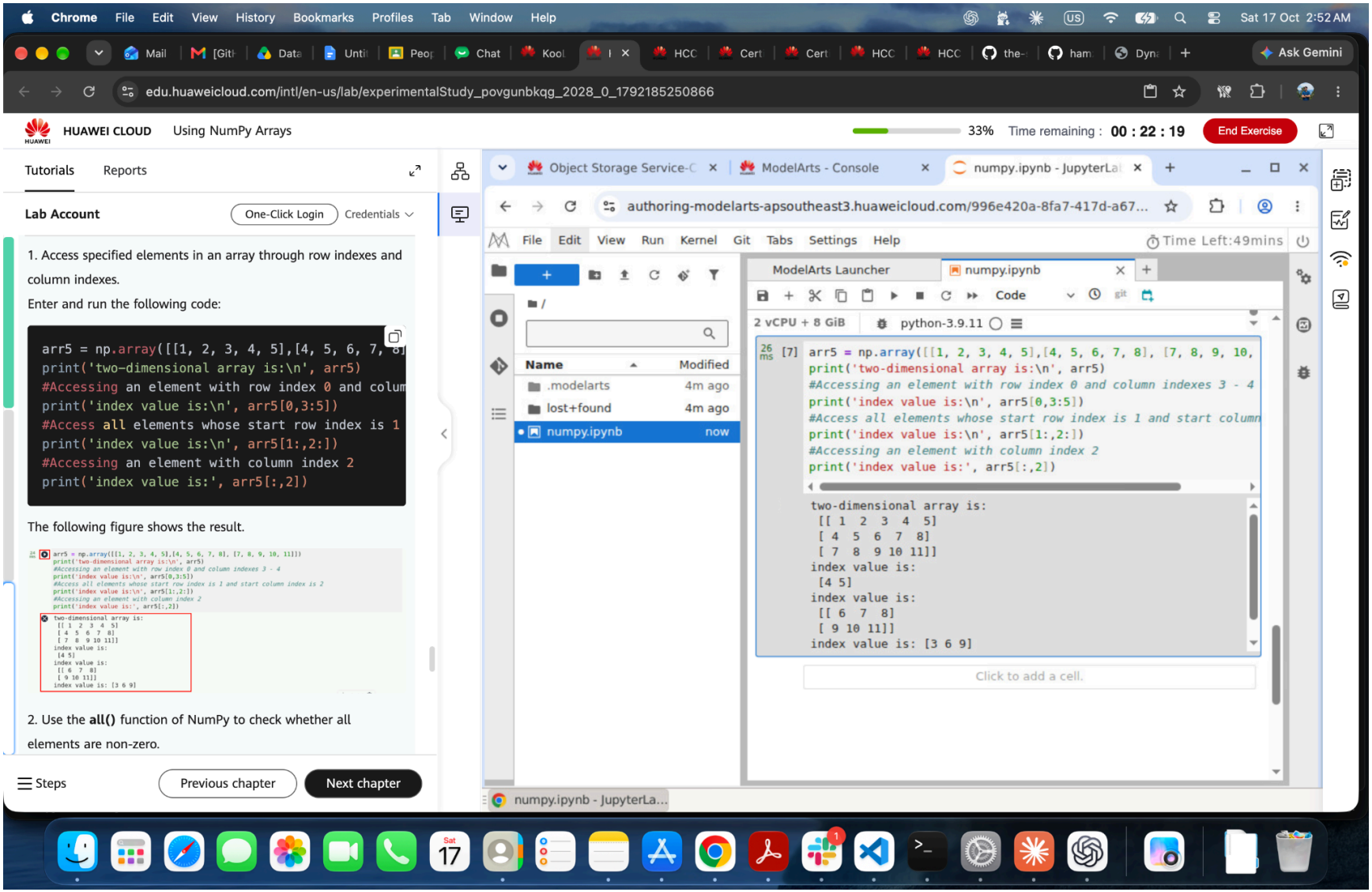

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 17 Oct 2:48 AM
edu.huaweicloud.com/intl/en-us/lab/experimentalStudy_povgunbkqg_2028_0_1792185250866
HUAWEI CLOUD
Using NumPy Arrays
33%
Time remaining : 00 : 26 : 05
End Exercise
Tutorials
Reports
Lab Account
One-Click Login
Credentials
np.save("/data/input/task_02_arr1", arr1)
The following figure shows the result.
Step 5 Defining a NumPy Multi-dimensional Array
1. Create a two-dimensional array.
Enter and run the following code:
#Create a two-dimensional array
arr2 = np.array([[1, 2, 3, 4],[4, 5, 6, 7], [7,
print('Array Created:\n',arr2)
The following figure shows the result.
Steps
Previous chapter
Next chapter
Object Storage Service-C
ModelArts - Console
numpy.ipynb - JupyterLa
authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...
File Edit View Run Kernel Git Tabs Settings Help
Time Left:54mins
ModelArts Launcher
numpy.ipynb
Code
2 vCPU + 8 GiB
python-3.9.11
Name
Modified
.modelarts
4m ago
lost+found
4m ago
numpy.ipynb
now
[1] #Import the NumPy library. The following experiments are perform
import numpy as np
#Create a one-dimensional array
arr1 = np.array([1, 2, 3, 4])
print('Array Created:',arr1)
np.save("/data/input/task_02_arr1", arr1)
Array Created: [1 2 3 4]
[2] #Create a two-dimensional array
arr2 = np.array([[1, 2, 3, 4],[4, 5, 6, 7], [7, 8, 9, 10]])
print('Array Created:\n',arr2)
Array Created:
[[1 2 3 4]
[4 5 6 7]
[7 8 9 10]]
Click to add a cell.
numpy.ipynb - JupyterLa...


HUAWEI CLOUD
Using NumPy Arrays
33%
Time remaining : 00 : 25 : 22
End Exercise
Tutorials
Reports
Lab Account
One-Click Login
Credentials
arr2 = np.array([[1, 2, 3, 4],[4, 5, 6, 7], [7,
print('Array Created:\n',arr2)
The following figure shows the result.
#Create a two-dimensional array
arr2 = np.array([[1, 2, 3, 4],[4, 5, 6, 7], [7, 8, 9, 10]])
print('Array Created: \n',arr2)
Array Created:
[[1 2 3 4]
[4 5 6 7]
[7 8 9 10]]
2. Create a three-dimensional array.
Enter and run the following code:
#Create a three-dimensional array
arr3 = np.array([[[1, 1], [2, 2], [3, 3], [4, 4]
[[4, 4], [5, 5], [6, 6], [7, 7]
[[7, 7], [8, 8], [9, 9], [10, 1
print('Array Created:\n',arr3)
The following figure shows the result.
#Create a three-dimensional array
arr3 = np.array([[[1, 1], [2, 2], [3, 3], [4, 4]],
[[4, 4], [5, 5], [6, 6], [7, 7]],
[[7, 7], [8, 8], [9, 9], [10, 10]]])
print('Array Created: \n',arr3)
Array Created:
Steps
Previous chapter
Next chapter
Object Storage Service-C
ModelArts - Console
numpy.ipynb - JupyterLa
authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...
File Edit View Run Kernel Git Tabs Settings Help
Time Left:52mins
ModelArts Launcher
numpy.ipynb
Code
2 vCPU + 8 GiB
Python-3.9.11
[7 8 9 10]]
Name
Modified
.modelarts
4m ago
lost+found
4m ago
numpy.ipynb
now
[3]
#Create a three-dimensional array
arr3 = np.array([[[1, 1], [2, 2], [3, 3], [4, 4]],
[[4, 4], [5, 5], [6, 6], [7, 7]],
[[7, 7], [8, 8], [9, 9], [10, 10]]])
print('Array Created:\n',arr3)
Array Created:
[[[1 1]
[2 2]
[3 3]
[4 4]]
[[4 4]
[5 5]
[6 6]
[7 7]]
[[7 7]
[8 8]
[9 9]
[10 10]]]
Click to add a cell.
numpy.ipynb - JupyterLa...


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 17 Oct 2:49 AM
edu.huaweicloud.com/intl/en-us/lab/experimentalStudy_povgunbkqg_2028_0_1792185250866
HUAWEI CLOUD
Using NumPy Arrays
33%
Time remaining : 00 : 24 : 34
End Exercise
Tutorials
Reports
Lab Account
One-Click Login
Credentials
[[[1 1]
[2 2]
[3 3]
[4 4]]
[[4 4]
[5 5]
[6 6]
[7 7]]
[[7 7]
[8 8]
[9 9]
[10 10]]]
3. View the array properties.
Enter and run the following code:
#View the dimension of an array
print("dimension of arr3 is:\n", arr3.ndim)
#View the shape of an array
print("shape of arr3 is:\n", arr3.shape)
#View the size of an array
print("size of arr3 is:\n", arr3.size)
#View the number of bytes occupied by an array e
print("itemsize of arr3 is:\n", arr3.itemsize)
The following figure shows the result.
Steps
Previous chapter
Next chapter
Object Storage Service-C
ModelArts - Console
numpy.ipynb - JupyterLa
authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...
File Edit View Run Kernel Git Tabs Settings Help
Time Left:52mins
ModelArts Launcher
numpy.ipynb
Code
2 vCPU + 8 Gi
python-3.9.11
[10 10]]]
Name
Modified
.modelarts 4m ago
lost+found 4m ago
numpy.ipynb now
[4]
#View the dimension of an array
print("dimension of arr3 is:\n", arr3.ndim)
#View the shape of an array
print("shape of arr3 is:\n", arr3.shape)
#View the size of an array
print("size of arr3 is:\n", arr3.size)
#View the number of bytes occupied by an array element.
print("itemsize of arr3 is:\n", arr3.itemsize)
dimension of arr3 is:
3
shape of arr3 is:
(3, 4, 2)
size of arr3 is:
24
itemsize of arr3 is:
8
Click to add a cell.
numpy.ipynb - JupyterLa...


HUAWEI CLOUD
Using NumPy Arrays
33%
Time remaining : 00 : 23 : 50
End Exercise
Tutorials
Reports
Lab Account
One-Click Login
Credentials
dimension of arr3 is:
3
shape of arr3 is:
(3, 4, 2)
size of arr3 is:
24
itemsize of arr3 is:
8
4. Use the functions provided by NumPy to create arrays.
Enter and run the following code:
print('array created by arange function is:\n',
print('array created by linspace function is:\n'
print('array created by logspace function is:\n'
print('array created by zeros function is:\n',
print('array created by eye function is:\n', np.
print('array created by diag function is:\n', np
np.save("/data/input/task_02_diag", np.diag([1,2
The following figure shows the result.
Steps
Previous chapter
Next chapter
authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...
Time Left:51mins
ModelArts Launcher
numpy.ipynb
Code
2 vCPU + 8 GiB
python-3.9.11
itemsize of arr3 is:
8
Name
Modified
.modelarts
4m ago
lost+found
4m ago
numpy.ipynb
now
print('array created by arange function is:\n', np.arange(0,1,0.
print('array created by linspace function is:\n', np.linspace(0,
print('array created by logspace function is:\n', np.logspace(0,
print('array created by zeros function is:\n', np.zeros((2,3)))
print('array created by eye function is:\n', np.eye(3))
print('array created by diag function is:\n', np.diag([1,2,3,4])
np.save("/data/input/task_02_diag", np.diag([1,2,3,4]))
array created by linspace function is:
[0. 0.09090909 0.18181818 0.27272727 0.36363636 0.45
454545
0.54545455 0.63636364 0.72727273 0.81818182 0.90909091 1.
]
array created by logspace function is:
[1. 1.27427499 1.62377674 2.06913808 2.636
6509
3.35981829 4.2813324 5.45559478 6.95192796 8.8586
679
11.28837892 14.38449888 18.32980711 23.35721469 29.7635
1442
37.92690191 48.32930239 61.58482111 78.47599704 100.
]
array created by zeros function is:
[[0. 0. 0.]
[0. 0. 0.]]
numpy.ipynb - JupyterLa...


HUAWEI CLOUD
Using NumPy Arrays
33%
Time remaining : 00 : 23 : 11
End Exercise
Tutorials
Reports
Lab Account
One-Click Login
Credentials
Step 6 Indexing of NumPy One-Dimensional Arrays
1. Access elements of one-dimensional arrays through indexing and slicing.
Enter and run the following code:
arr4 = np.arange(10)
#Accessing the element with index 5
print('index value is:\n', arr4[5])
#Accessing the element with index 3-4
print('index value is:\n',arr4[3:5])
#Accessing the element with index 0-4
print('index value is:',arr4[:5])
#Access to the last element
print('index value is:',arr4[-1])
The following figure shows the result.
Steps
Previous chapter
Next chapter
ModelArts Launcher
numpy.ipynb
index value is:
5
index value is:
[3 4]
index value is: [0 1 2 3 4]
index value is: 9
Click to add a cell.


Using NumPy Arrays
33%
Time remaining : 00 : 22 : 19
End Exercise
Tutorials
Reports
Lab Account
One-Click Login
Credentials
1. Access specified elements in an array through row indexes and column indexes.
Enter and run the following code:
arr5 = np.array([[1, 2, 3, 4, 5],[4, 5, 6, 7, 8]
print('two-dimensional array is:\n', arr5)
#Accessing an element with row index 0 and colum
print('index value is:\n', arr5[0,3:5])
#Access all elements whose start row index is 1
print('index value is:\n', arr5[1:,2:])
#Accessing an element with column index 2
print('index value is:', arr5[:,2])
The following figure shows the result.
two-dimensional array is:
[[1 2 3 4 5]
[4 5 6 7 8]
[7 8 9 10 11]]
index value is:
[4 5]
index value is:
[[6 7 8]
[9 10 11]]
index value is: [3 6 9]
2. Use the all() function of NumPy to check whether all elements are non-zero.
Steps
Previous chapter
Next chapter
Object Storage Service-C
ModelArts - Console
numpy.ipynb - JupyterLa
authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...
File Edit View Run Kernel Git Tabs Settings Help
Time Left:49mins
ModelArts Launcher
numpy.ipynb
Code
2 vCPU + 8 GiB
python-3.9.11
Name
Modified
.modelarts
4m ago
lost+found
4m ago
numpy.ipynb
now
[7]
arr5 = np.array([[1, 2, 3, 4, 5],[4, 5, 6, 7, 8], [7, 8, 9, 10,
print('two-dimensional array is:\n', arr5)
#Accessing an element with row index 0 and column indexes 3 - 4
print('index value is:\n', arr5[0,3:5])
#Access all elements whose start row index is 1 and start column
print('index value is:\n', arr5[1:,2:])
#Accessing an element with column index 2
print('index value is:', arr5[:,2])
two-dimensional array is:
[[1 2 3 4 5]
[4 5 6 7 8]
[7 8 9 10 11]]
index value is:
[4 5]
index value is:
[[6 7 8]
[9 10 11]]
index value is: [3 6 9]
Click to add a cell.
numpy.ipynb - JupyterLa...

HUAWEI CLOUD Using NumPy Arrays

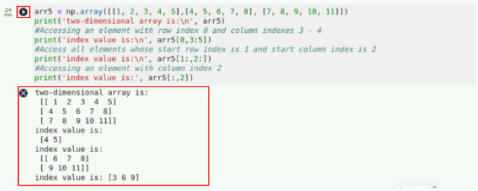
33% Time remaining : 00 : 21 : 47 End Exercise

Tutorials Reports

Lab Account One-Click Login Credentials

```
print('index value is:\n', arr5[0,3:5])
#Access all elements whose start row index is 1
print('index value is:\n', arr5[1:,2:])
#Accessing an element with column index 2
print('index value is:', arr5[:,2])
```

The following figure shows the result.



2. Use the **all()** function of NumPy to check whether all elements are non-zero.

Enter and run the following code:

```
#np.all() Returns True if all elements are non-z
arr6 = np.array([[1, -1, 3, 4, 5],[4, 5, 6, 7, 8
print(np.all(arr6))
```

The following figure shows the result.

Steps Previous chapter Next chapter

authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...

Time Left:49mins

```
#np.all() Returns True if all elements are non-zero
arr6 = np.array([[1, -1, 3, 4, 5],[4, 5, 6, 7, 8], [7, 8, 9, 10,
print(np.all(arr6))
```

True

Click to add a cell.

HUAWEI CLOUD Using NumPy Arrays

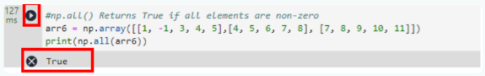
33% Time remaining : 00 : 21 : 06 End Exercise

Tutorials Reports

Lab Account One-Click Login Credentials

```
print(np.all(arr6))
```

The following figure shows the result.

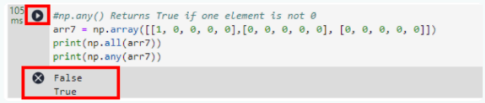


3. Use the **any()** function of NumPy to check whether an array contains non-zero elements.

Enter and run the following code:

```
#np.any() Returns True if one element is not 0
arr7 = np.array([[1, 0, 0, 0, 0],[0, 0, 0, 0, 0]
print(np.all(arr7))
print(np.any(arr7))
```

The following figure shows the result.



4. Use the **where** function of NumPy to find the elements that meet the conditions.

Steps Previous chapter Next chapter

authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...

File Edit View Run Kernel Git Tabs Settings Help Time Left:48mins

ModelArts Launcher numpy.ipynb

Code

2 vCPU + 8 GiB python-3.9.11

Name Modified

.modelarts 4m ago

lost+found 4m ago

numpy.ipynb now

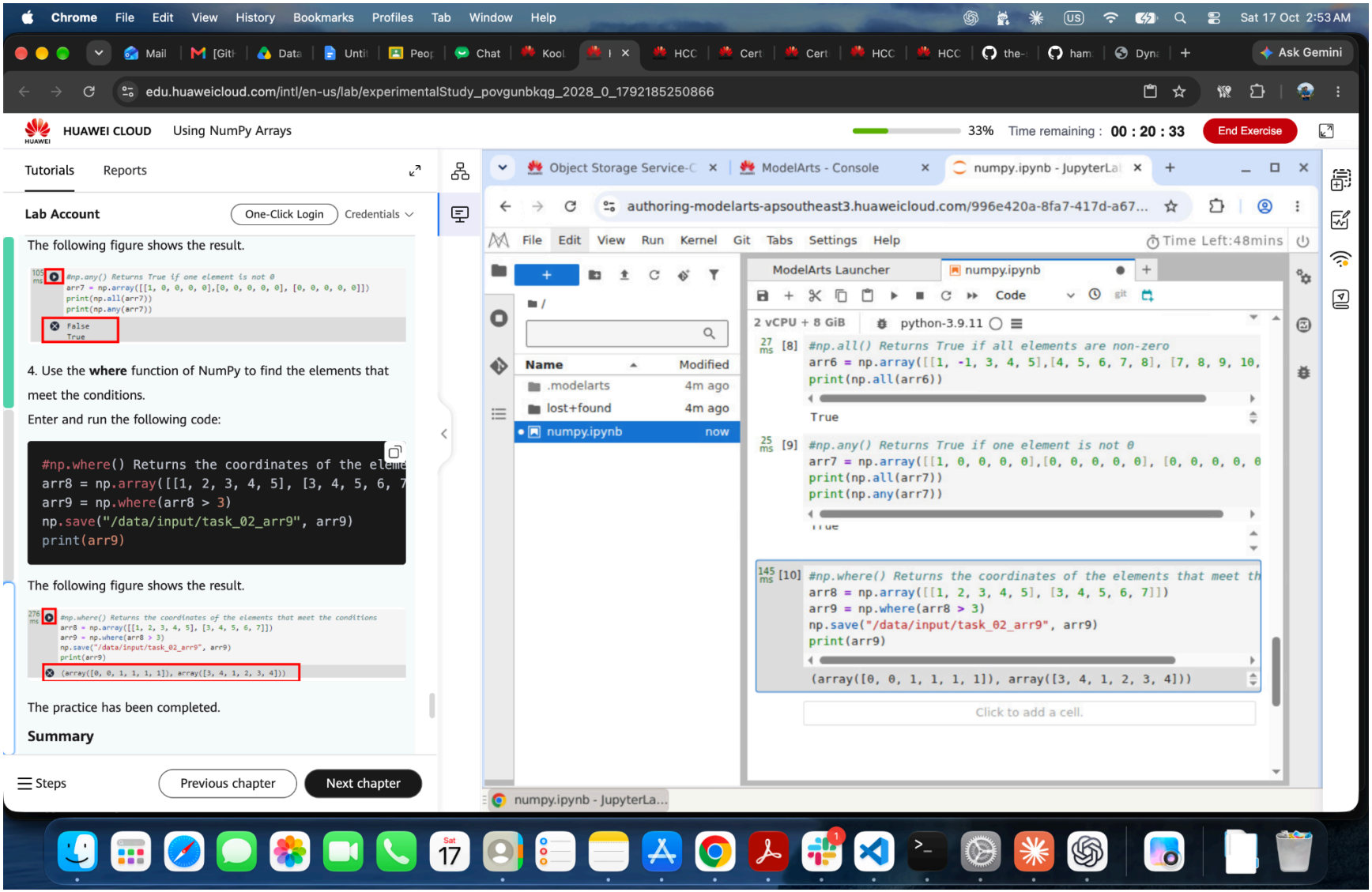
```
[8] #np.all() Returns True if all elements are non-zero
arr6 = np.array([[1, -1, 3, 4, 5],[4, 5, 6, 7, 8], [7, 8, 9, 10,
print(np.all(arr6))
```

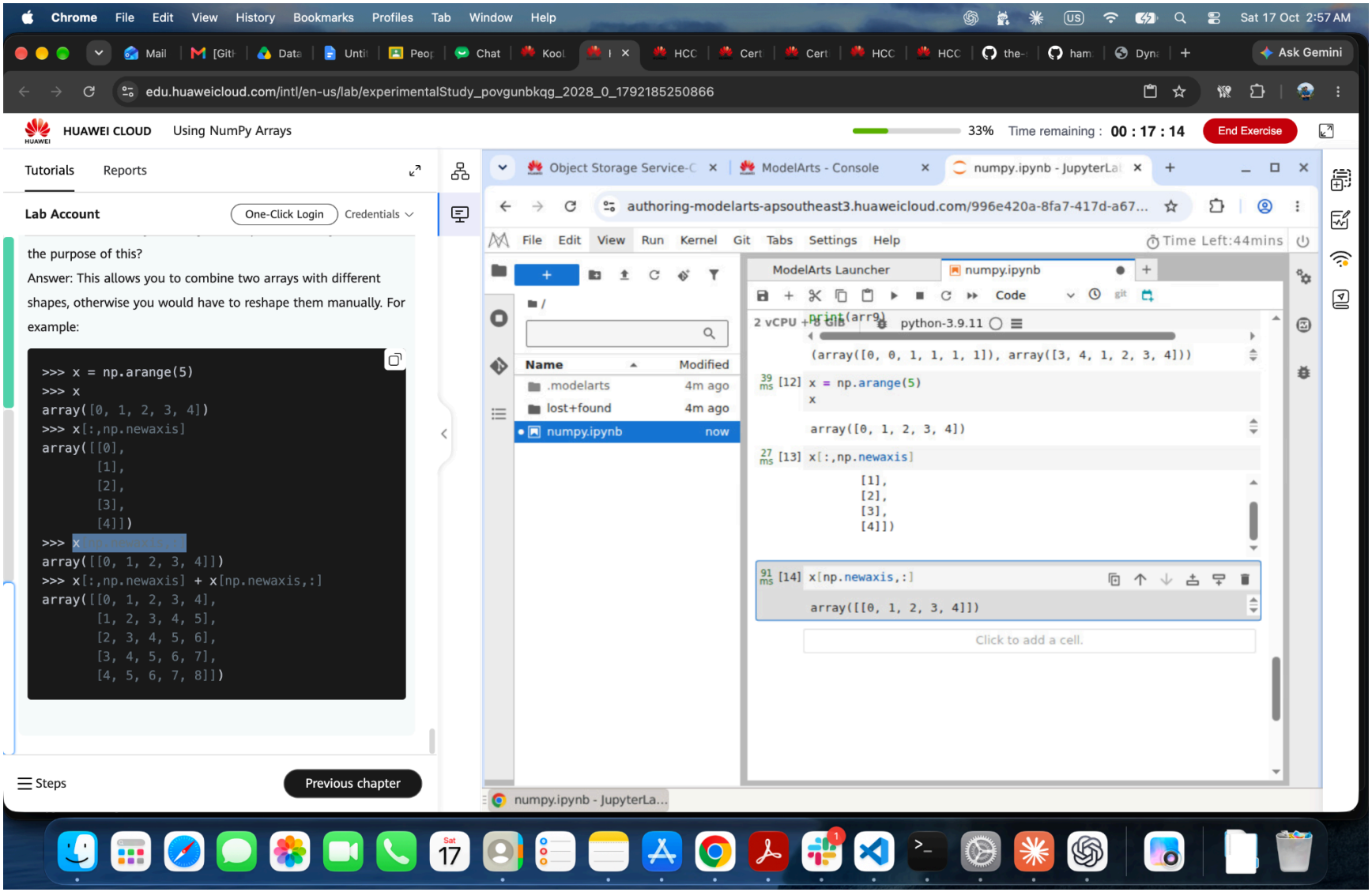
True

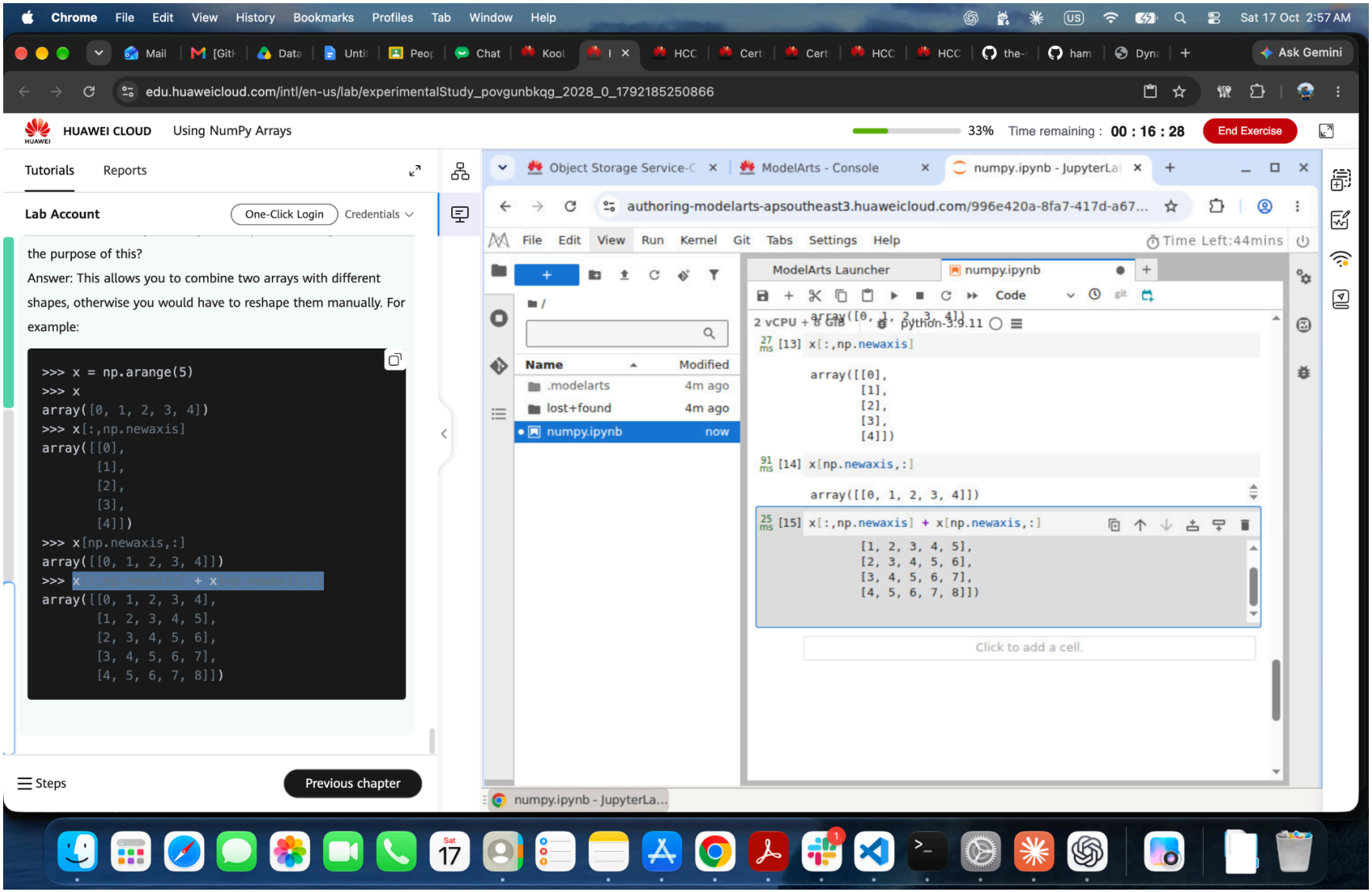
```
[9] #np.any() Returns True if one element is not 0
arr7 = np.array([[1, 0, 0, 0, 0],[0, 0, 0, 0, 0], [0, 0, 0, 0, 0
print(np.all(arr7))
print(np.any(arr7))
```

False
True

Click to add a cell.


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 17 Oct 2:53 AM
edu.huaweicloud.com/intl/en-us/lab/experimentalStudy_povgunbkqg_2028_0_1792185250866
HUAWEI CLOUD
Using NumPy Arrays
33%
Time remaining : 00 : 20 : 33
End Exercise
Tutorials
Reports
Lab Account
One-Click Login
Credentials
The following figure shows the result.
#np.any() Returns True if one element is not 0
arr7 = np.array([[1, 0, 0, 0, 0],[0, 0, 0, 0, 0], [0, 0, 0, 0, 0]])
print(np.all(arr7))
print(np.any(arr7))
False
True
4. Use the where function of NumPy to find the elements that meet the conditions.
Enter and run the following code:
#np.where() Returns the coordinates of the eleme
arr8 = np.array([[1, 2, 3, 4, 5], [3, 4, 5, 6, 7
arr9 = np.where(arr8 > 3)
np.save("/data/input/task_02_arr9", arr9)
print(arr9)
The following figure shows the result.
#np.where() Returns the coordinates of the elements that meet the conditions
arr8 = np.array([[1, 2, 3, 4, 5], [3, 4, 5, 6, 7]])
arr9 = np.where(arr8 > 3)
np.save("/data/input/task_02_arr9", arr9)
print(arr9)
(array([0, 0, 1, 1, 1, 1]), array([3, 4, 1, 2, 3, 4]))
The practice has been completed.
Summary
Steps
Previous chapter
Next chapter
Object Storage Service-C
ModelArts - Console
numpy.ipynb - JupyterLa
authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...
File Edit View Run Kernel Git Tabs Settings Help
Time Left:48mins
ModelArts Launcher
numpy.ipynb
Code
2 vCPU + 8 GiB
python-3.9.11
Name
Modified
.modelarts
4m ago
lost+found
4m ago
numpy.ipynb
now
[8] #np.all() Returns True if all elements are non-zero
arr6 = np.array([[1, -1, 3, 4, 5],[4, 5, 6, 7, 8], [7, 8, 9, 10,
print(np.all(arr6))
True
[9] #np.any() Returns True if one element is not 0
arr7 = np.array([[1, 0, 0, 0, 0],[0, 0, 0, 0, 0], [0, 0, 0, 0, 0
print(np.all(arr7))
print(np.any(arr7))
[10] #np.where() Returns the coordinates of the elements that meet th
arr8 = np.array([[1, 2, 3, 4, 5], [3, 4, 5, 6, 7]])
arr9 = np.where(arr8 > 3)
np.save("/data/input/task_02_arr9", arr9)
print(arr9)
(array([0, 0, 1, 1, 1, 1]), array([3, 4, 1, 2, 3, 4]))
Click to add a cell.
numpy.ipynb - JupyterLa...


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 17 Oct 2:57 AM
edu.huaweicloud.com/intl/en-us/lab/experimentalStudy_povgunbkqg_2028_0_1792185250866
HUAWEI CLOUD
Using NumPy Arrays
33%
Time remaining : 00 : 17 : 14
End Exercise
Tutorials
Reports
Lab Account
One-Click Login
Credentials
the purpose of this?
Answer: This allows you to combine two arrays with different shapes, otherwise you would have to reshape them manually. For example:
>>> x = np.arange(5)
>>> x
array([0, 1, 2, 3, 4])
>>> x[:,np.newaxis]
array([[0],
[1],
[2],
[3],
[4]])
>>> x[np.newaxis,:]
array([[0, 1, 2, 3, 4]])
>>> x[:,np.newaxis] + x[np.newaxis,:]
array([[0, 1, 2, 3, 4],
[1, 2, 3, 4, 5],
[2, 3, 4, 5, 6],
[3, 4, 5, 6, 7],
[4, 5, 6, 7, 8]])
Steps
Previous chapter
Object Storage Service-C
ModelArts - Console
numpy.ipynb - JupyterLa
authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...
File Edit View Run Kernel Git Tabs Settings Help
Time Left:44mins
ModelArts Launcher
numpy.ipynb
Code
2 vCPU + 8 GiB
python-3.9.11
Name
Modified
.modelarts
4m ago
lost+found
4m ago
numpy.ipynb
now
print(arr9)
(array([0, 0, 1, 1, 1, 1]), array([3, 4, 1, 2, 3, 4]))
[12] x = np.arange(5)
x
array([0, 1, 2, 3, 4])
[13] x[:,np.newaxis]
[1],
[2],
[3],
[4]])
[14] x[np.newaxis,:]
array([[0, 1, 2, 3, 4]])
Click to add a cell.
numpy.ipynb - JupyterLa...


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 17 Oct 2:57 AM
edu.huaweicloud.com/intl/en-us/lab/experimentalStudy_povgunbkqg_2028_0_1792185250866
HUAWEI CLOUD
Using NumPy Arrays
33%
Time remaining : 00 : 16 : 28
End Exercise
Tutorials
Reports
Lab Account
One-Click Login
Credentials
the purpose of this?
Answer: This allows you to combine two arrays with different shapes, otherwise you would have to reshape them manually. For example:
>>> x = np.arange(5)
>>> x
array([0, 1, 2, 3, 4])
>>> x[:,np.newaxis]
array([[0],
[1],
[2],
[3],
[4]])
>>> x[np.newaxis,:]
array([[0, 1, 2, 3, 4]])
>>> x[:,np.newaxis] + x[np.newaxis,:]
array([[0, 1, 2, 3, 4],
[1, 2, 3, 4, 5],
[2, 3, 4, 5, 6],
[3, 4, 5, 6, 7],
[4, 5, 6, 7, 8]])
Steps
Previous chapter
Object Storage Service-C
ModelArts - Console
numpy.ipynb - JupyterLa
authoring-modelarts-apsoutheast3.huaweicloud.com/996e420a-8fa7-417d-a67...
File Edit View Run Kernel Git Tabs Settings Help
Time Left:44mins
ModelArts Launcher
numpy.ipynb
Code
2 vCPU + 8 GiB
python-3.9.11
Name
Modified
.modelarts 4m ago
lost+found 4m ago
numpy.ipynb now
[13] x[:,np.newaxis]
array([[0],
[1],
[2],
[3],
[4]])
[14] x[np.newaxis,:]
array([[0, 1, 2, 3, 4]])
[15] x[:,np.newaxis] + x[np.newaxis,:]
[1, 2, 3, 4, 5],
[2, 3, 4, 5, 6],
[3, 4, 5, 6, 7],
[4, 5, 6, 7, 8]])
Click to add a cell.
numpy.ipynb - JupyterLa...